

EDUGENIE – AI POWERED LEARNING ASSISTANT :

ABSTRACT:

EduGenie is an AI-powered educational assistant developed to provide an interactive and personalized learning experience for students. The application uses the Google Gemini API to generate intelligent responses for educational queries. It enables students to ask questions, understand difficult concepts, summarize lengthy notes, generate quizzes, receive personalized learning recommendations, and summarize uploaded PDF documents. The system is developed using FastAPI for the backend and HTML, CSS, and JavaScript for the frontend. Additional features such as voice input, text-to-speech, dark mode, response download, and responsive design improve accessibility and user experience. EduGenie aims to simplify learning by providing instant AI-powered assistance through an intuitive web interface.

INTRODUCTION:

Artificial Intelligence has significantly transformed the education sector by providing intelligent learning tools that improve students' understanding of academic concepts. Traditional learning methods often require students to spend considerable time searching for study materials and understanding complex topics. AI-powered educational assistants overcome these challenges by offering instant explanations, summaries, quizzes, and personalized guidance.

EduGenie is designed as an intelligent learning assistant that integrates Google Gemini AI to assist students in learning efficiently. The application provides multiple educational services within a single platform, helping users improve productivity and knowledge acquisition.

PROBLEM STATEMENT:

Students frequently encounter challenges while understanding difficult concepts, summarizing lengthy notes, preparing quizzes, and organizing study plans. Existing educational platforms often provide limited interactive features and require switching between multiple applications.

Therefore, there is a need for an integrated AI-powered educational assistant that can provide intelligent answers, summarize content, generate quizzes, recommend learning paths, and assist students through voice interaction and PDF summarization.

OBJECTIVES:

The primary objectives of EduGenie are:

- Develop an AI-powered educational assistant.
- Provide intelligent question answering.
- Explain difficult concepts in simple language.

- Generate concise summaries from large text.
- Automatically create quizzes for revision.
- Recommend personalized learning paths.
- Summarize uploaded PDF documents.
- Support voice-based interaction.
- Improve accessibility using text-to-speech.
- Provide a modern and responsive user interface.

SCOPE OF THE PROJECT:

EduGenie is designed for a wide range of users, including:

- School Students
- College Students
- Competitive Exam Aspirants
- Teachers
- Self-Learners

The application can be used for:

- Learning new topics
- Understanding difficult concepts
- Preparing for examinations
- Summarizing study materials
- Creating quizzes
- Receiving AI-powered educational guidance

EXISTING SYSTEM:

Existing educational platforms generally provide only one or two learning services. Students often need multiple applications for summarization, note generation, quizzes, and AI assistance. Many systems lack personalized learning recommendations and integrated AI features.

PROPOSED SYSTEM:

EduGenie integrates multiple AI-powered educational services into one platform.

The proposed system uses FastAPI as the backend framework and Google Gemini AI for intelligent response generation. The application provides an interactive and responsive web interface developed using HTML, CSS, and JavaScript.

Additional features such as voice input, text-to-speech, PDF summarization, downloadable responses, dark mode, and responsive design significantly improve user experience.

FEATURES OF EDUGENIE:

- ☐ AI Question Answering
- ☐ Concept Explanation
- ☐ Smart Summarization
- ☐ Quiz Generation
- ☐ Personalized Learning Path
- ☐ PDF Upload and Summarization
- ☐ Voice Input
- ☐ Text-to-Speech
- ☐ Copy Response
- ☐ Download Response as PDF
- ☐ Download Response as TXT
- ☐ Dark Mode
- ☐ Responsive Design
- ☐ Typing Animation

HARDWARE REQUIREMENTS

- Processor:
Intel Core i3/i5 or above
- RAM:
Minimum 4 GB
- Storage:
10 GB Free Space
- Internet:
Broadband Internet Connection

SOFTWARE REQUIREMENTS:

Operating System

- Windows 10
- **Windows 11**

Programming Language

- Python 3.11

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- FastAPI

Artificial Intelligence

- **Google Gemini API**

Libraries

- Uvicorn
- python-dotenv
- PyPDF
- jsPDF

Development Tools

- Visual Studio Code
- Git
- GitHub

TECHNOLOGIES USED:

HTML:

HTML provides the structure of the web application and defines the layout of the user interface.

CSS:

CSS is used to create an attractive, responsive, and modern user interface.

JavaScript:

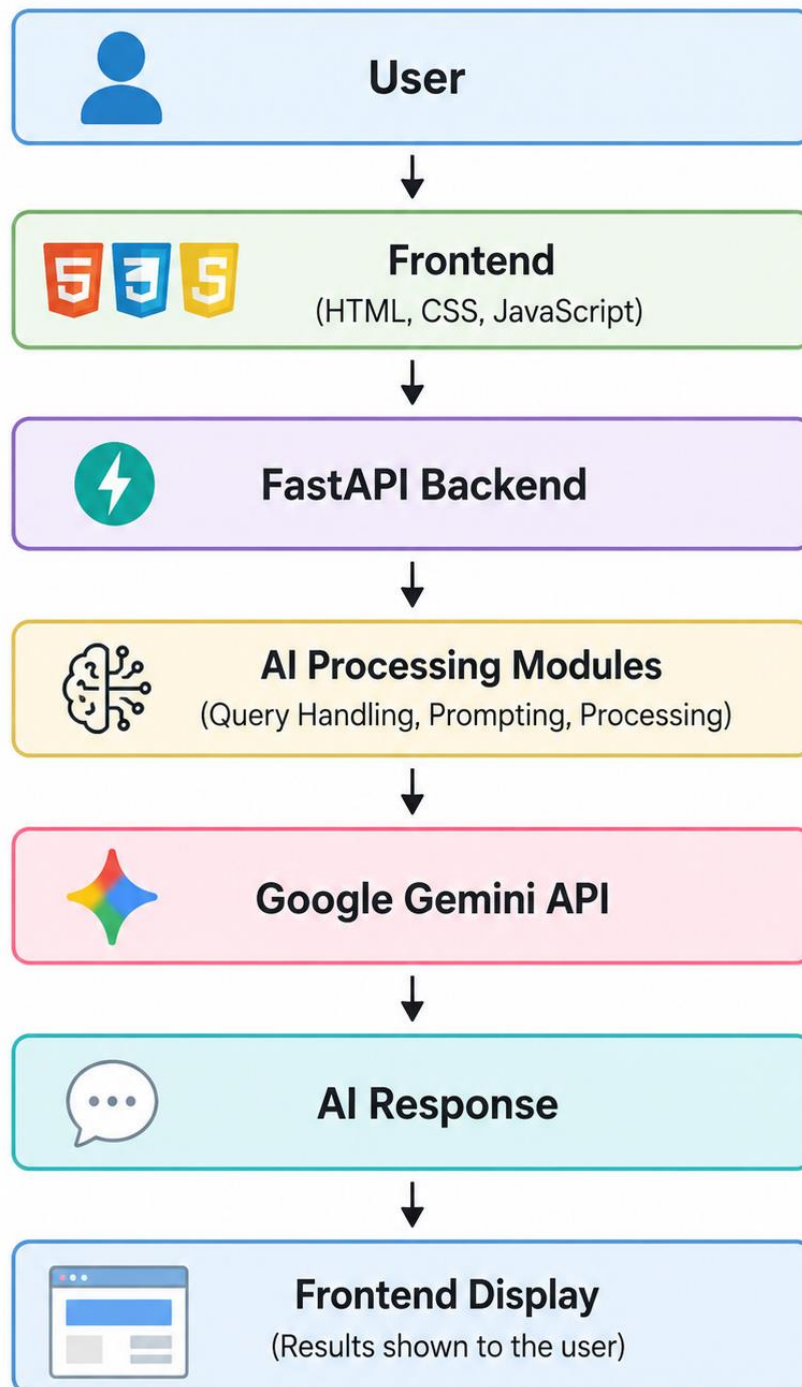
JavaScript provides dynamic functionality such as API calls, voice input, text-to-speech, downloading responses, and animations.

FastAPI :FastAPI is a high-performance Python framework used to build REST APIs that process user requests.

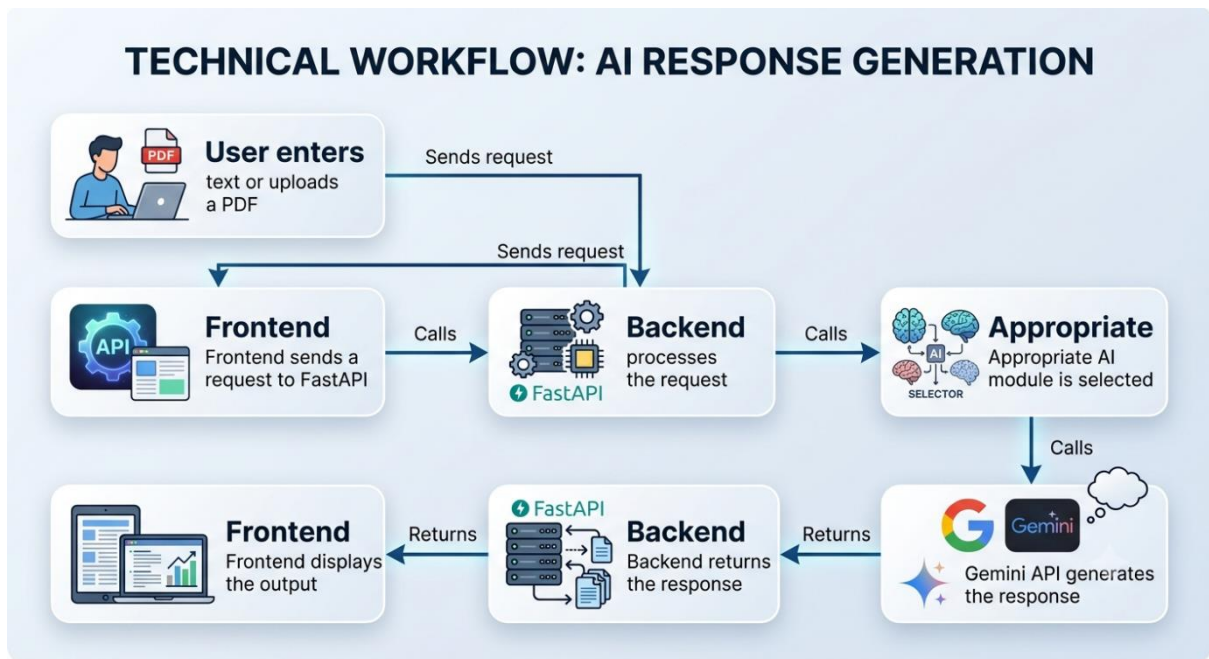
Google Gemini API:Google Gemini API generates AI-powered educational responses based on user input.

GitHub:GitHub is used for version control and project hosting.

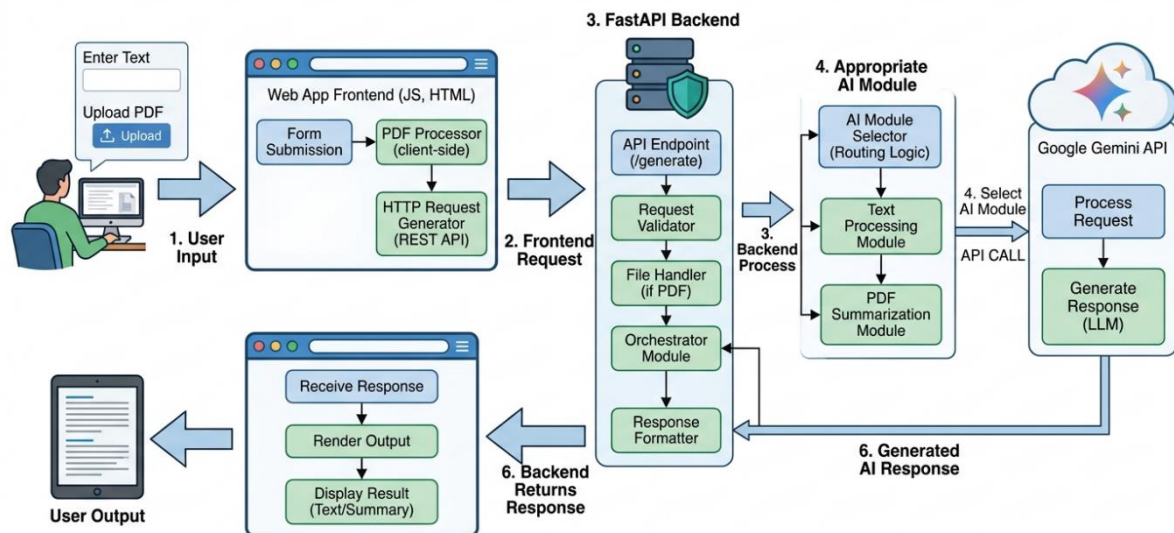
SYSTEM ARCHITECTURE:



SYSTEM WORKFLOW:



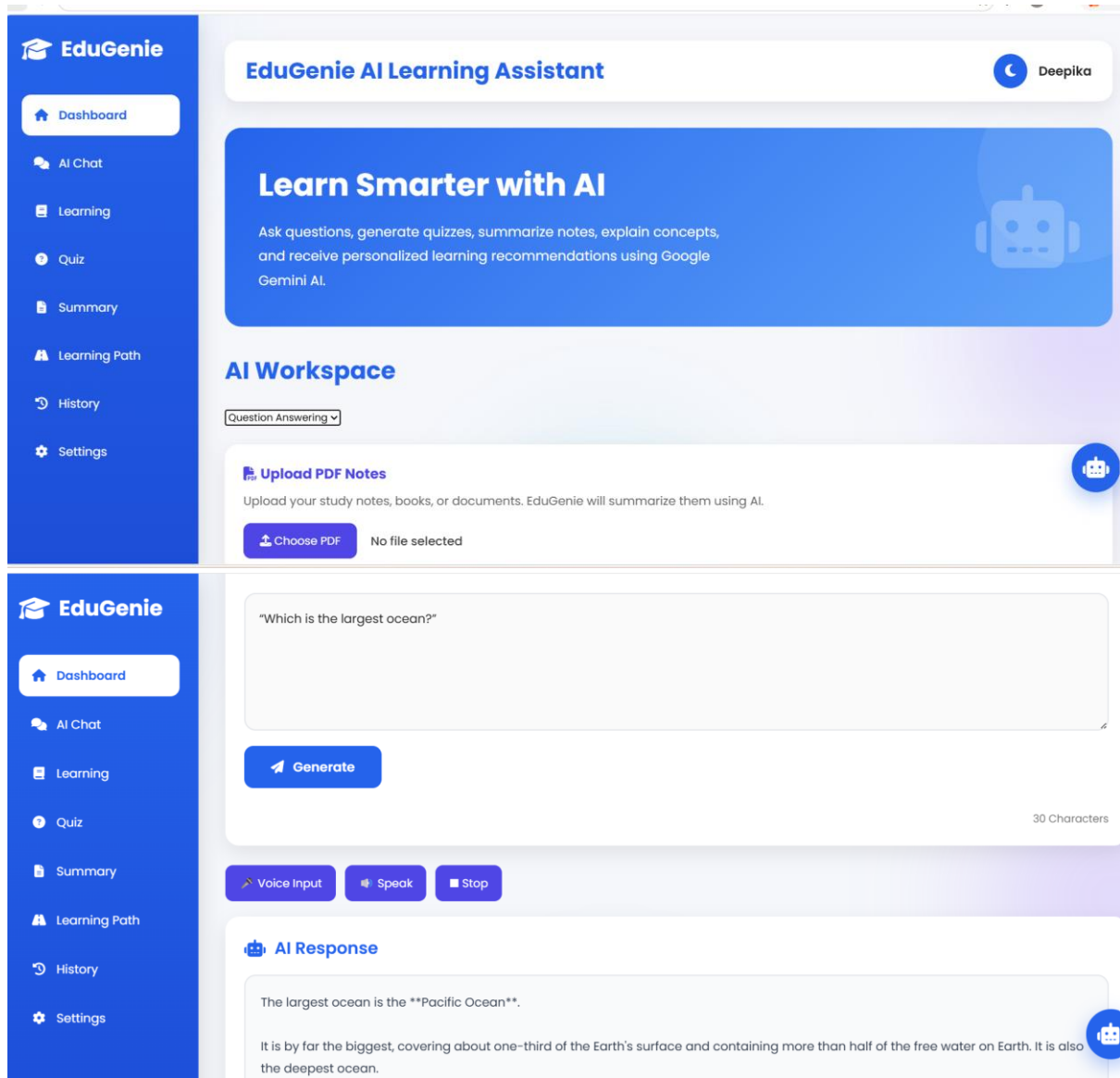
Generative AI Application Workflow Pipeline: Text & PDF Processing with FastAPI and Google Gemini API



MODULES:

Question Answering

This module allows users to ask educational questions and receive AI-generated responses using Google Gemini.



Concept Explanation

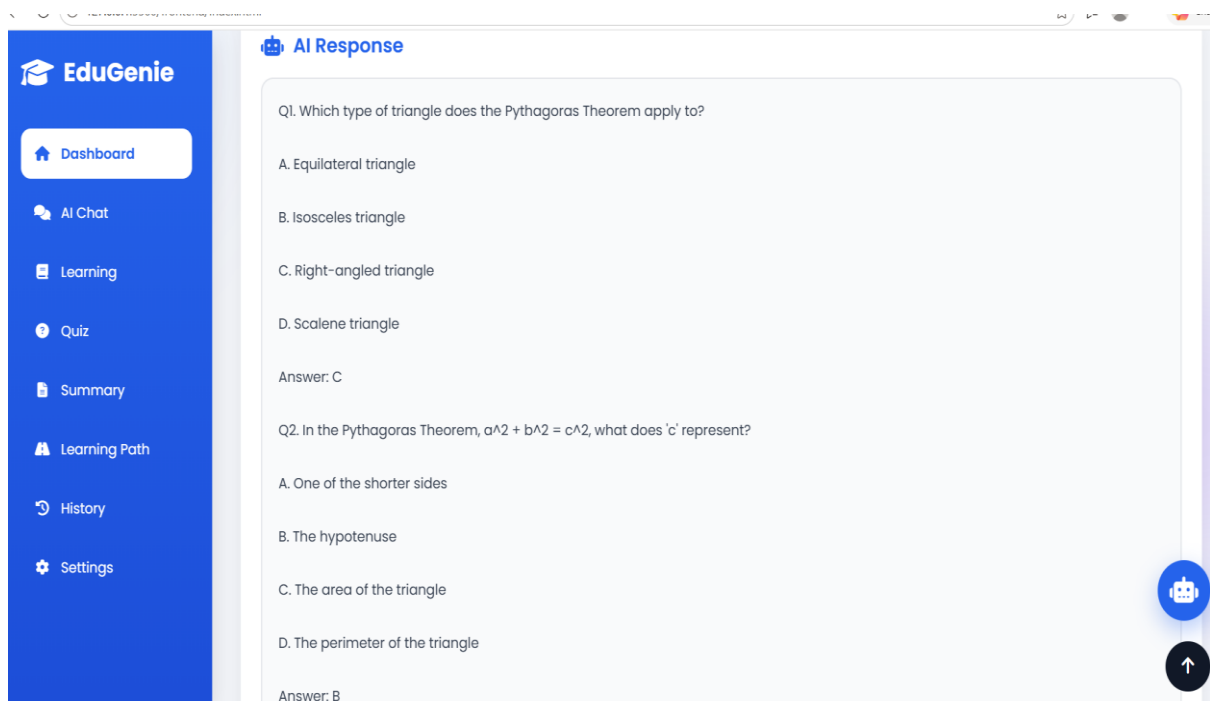
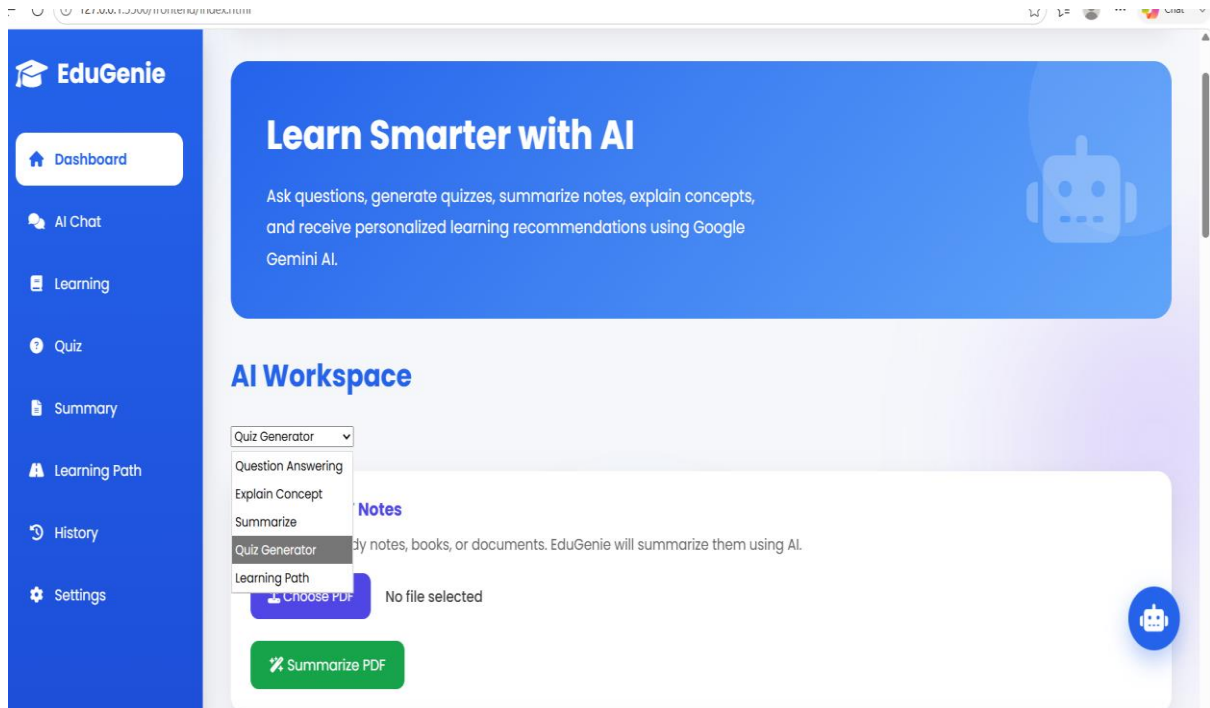
This module explains difficult concepts in simple and understandable language suitable for students.

Text Summarization

Users can paste lengthy notes or articles and obtain concise summaries.

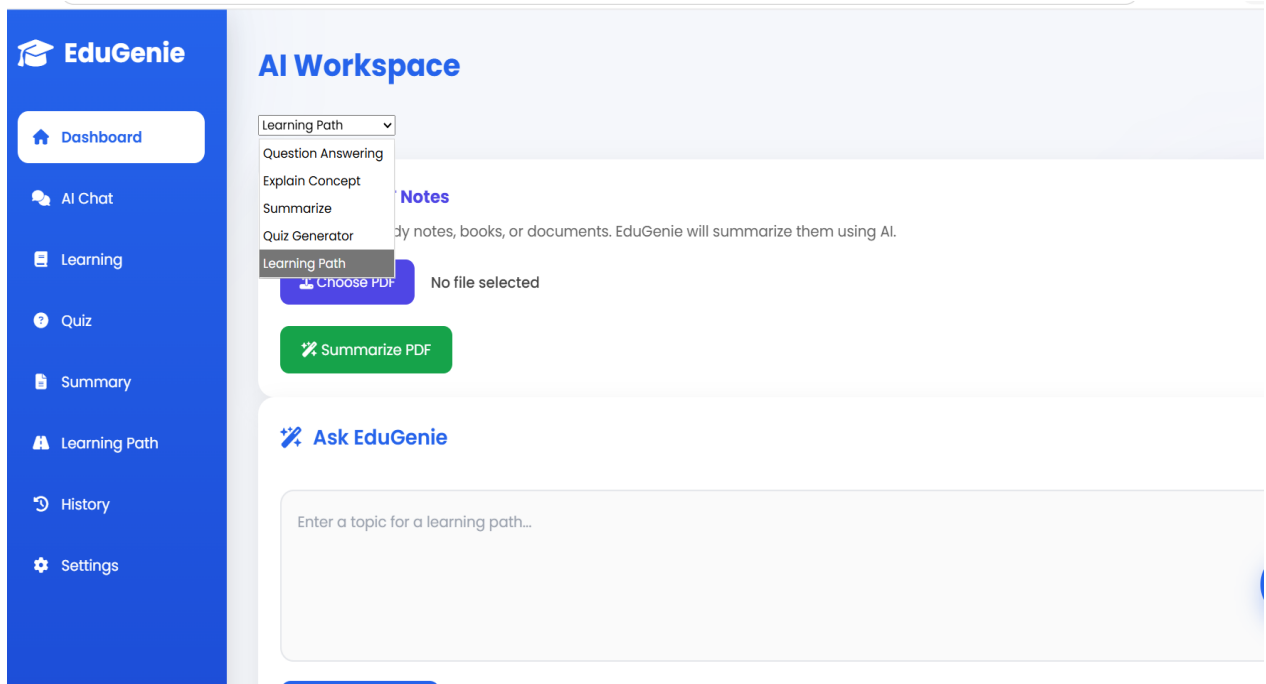
Quiz Generator

Automatically creates multiple-choice questions based on user-provided topics.

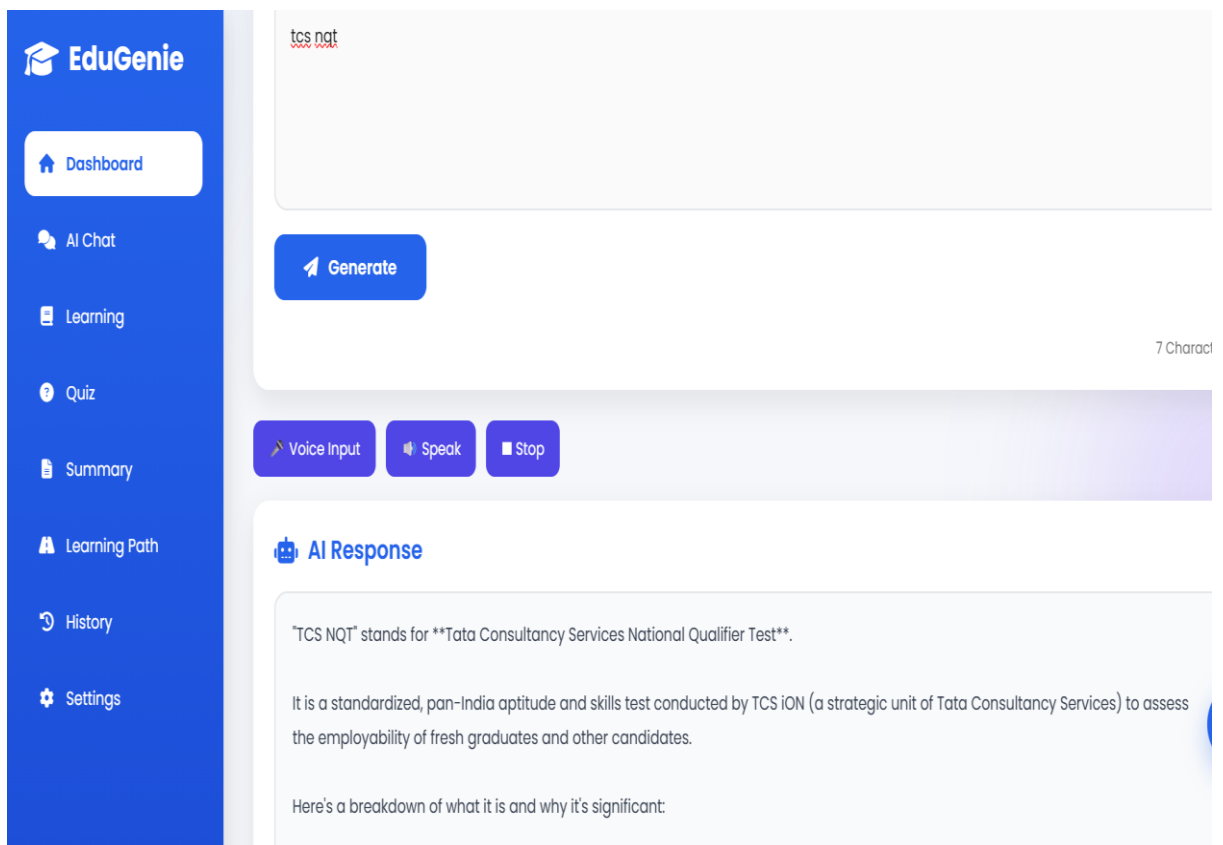


Learning Path Recommendation

Suggests personalized learning paths based on the selected topic.



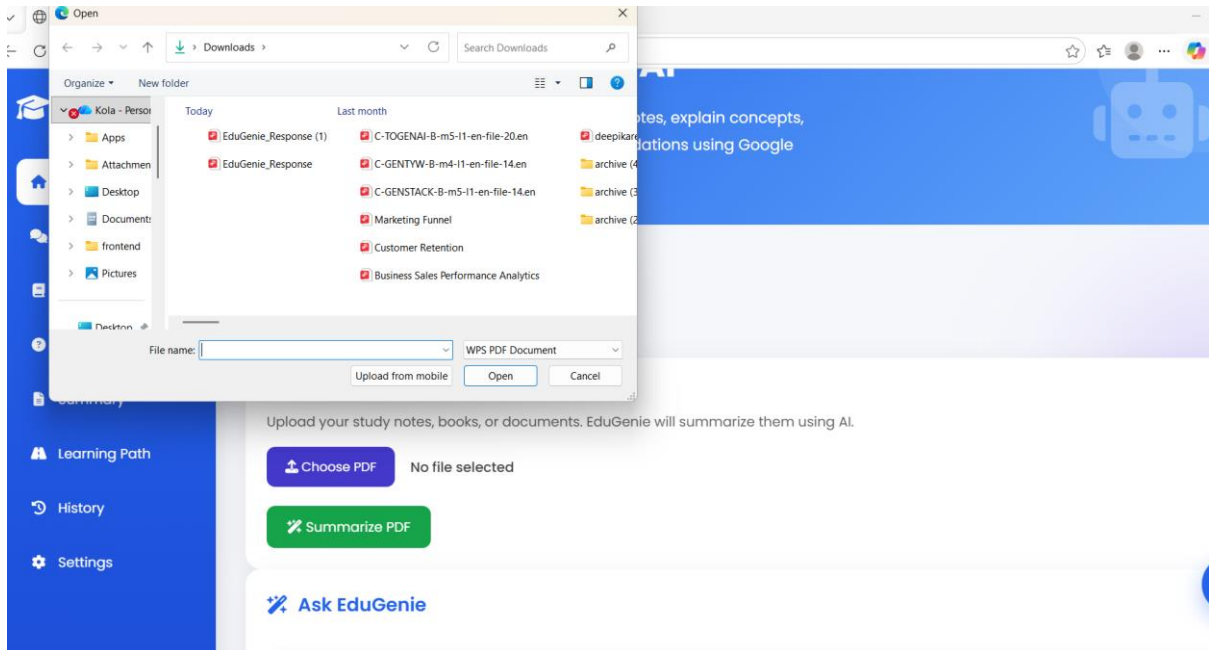
The screenshot shows the EduGenie AI Workspace interface. On the left is a blue sidebar with the EduGenie logo and a list of navigation options: Dashboard, AI Chat, Learning, Quiz, Summary, Learning Path, History, and Settings. The main area is titled "AI Workspace". A dropdown menu is open for the "Learning Path" option, showing a list of actions: Question Answering, Explain Concept, Summarize, Quiz Generator, and Learning Path. Below the dropdown, there is a "Choose PDF" button and a "No file selected" message. A green "Summarize PDF" button is also visible. To the right, there is a "Notes" section with a text area and a "Summarize" button. Below this is an "Ask EduGenie" section with a text input field labeled "Enter a topic for a learning path..." and a "Generate" button.



The screenshot shows the EduGenie AI Workspace interface. On the left is a blue sidebar with the EduGenie logo and a list of navigation options: Dashboard, AI Chat, Learning, Quiz, Summary, Learning Path, History, and Settings. The main area is titled "AI Workspace". At the top, there is a text input field containing "tcs nqt". Below this is a blue "Generate" button. To the right of the button, there is a character count "7 Character". Below the input field, there are three buttons: "Voice Input", "Speak", and "Stop". Below these buttons is an "AI Response" section. The response text reads: "TCS NQT" stands for **Tata Consultancy Services National Qualifier Test**. It is a standardized, pan-India aptitude and skills test conducted by TCS ION (a strategic unit of Tata Consultancy Services) to assess the employability of fresh graduates and other candidates. Here's a breakdown of what it is and why it's significant:

PDF Upload & Summarization

Allows users to upload PDF documents. The system extracts text and generates concise summaries using AI.



Voice Input

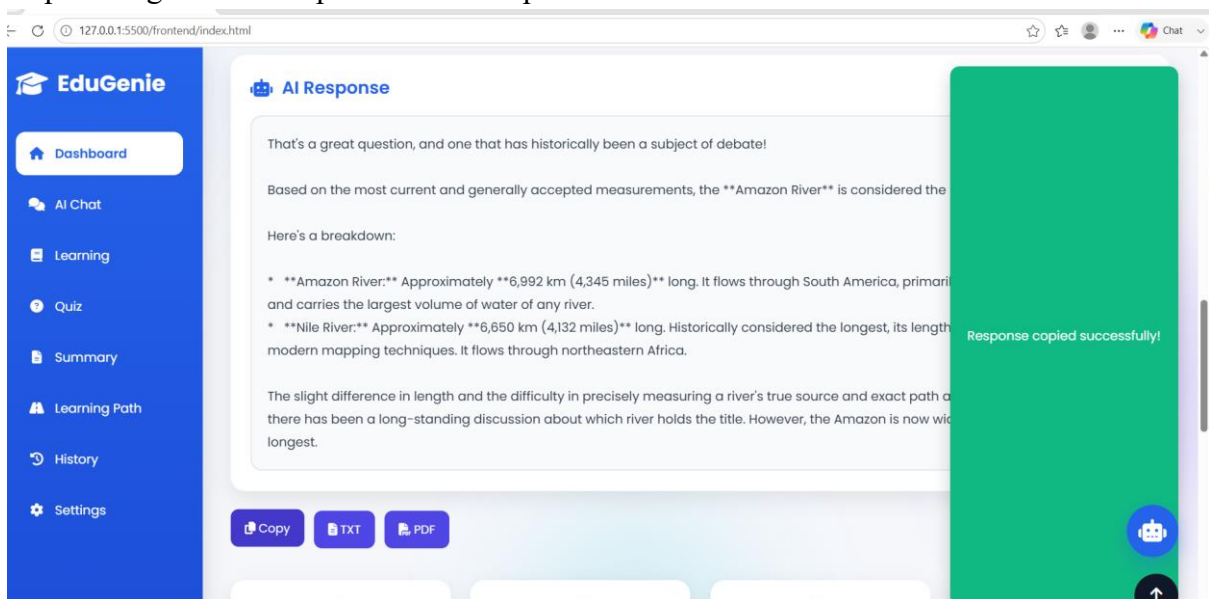
Users can ask questions using voice through the browser's Speech Recognition API.

Text-to-Speech

Reads AI-generated responses aloud to improve accessibility.

Copy Response

Copies AI-generated responses to the clipboard with one click.



Download Response

Users can download AI responses in PDF or TXT format.

PROJECT IMPLEMENTATION:

The implementation of EduGenie follows a modular client-server architecture, where the frontend and backend are developed independently and communicate through RESTful APIs. The frontend provides an intuitive interface for user interaction, while the backend handles request processing, AI integration, and response generation. Google Gemini API is used as the core AI engine to generate intelligent educational responses.

The application is designed to be scalable, responsive, and user-friendly, ensuring that students can access AI-powered learning assistance from any modern web browser.

Frontend Implementation:

The frontend is developed using **HTML5, CSS3, and JavaScript** to create a responsive and interactive user interface. HTML is used to structure the web pages, CSS is responsible for styling and responsive layouts, and JavaScript manages user interactions and communication with the backend APIs.

The interface includes multiple educational tools such as Question Answering, Concept Explanation, Summarization, Quiz Generation, Learning Path Recommendation, and PDF Upload. Users can select the desired module, provide input, and receive AI-generated responses instantly.

Additional frontend features include:

- Responsive design for desktop and mobile devices
- Dark/Light mode toggle
- Voice input using the Web Speech API
- Text-to-Speech for reading AI responses aloud
- Loading animations during API requests
- Typing animation for AI responses
- Copy response functionality
- Download responses as PDF and TXT files

These features improve usability and create an engaging learning experience.

Backend Implementation:

The backend is developed using **FastAPI**, a modern Python framework known for its high performance and automatic API documentation. FastAPI efficiently processes user requests, validates input data, communicates with AI modules, and returns structured JSON responses.

The backend follows a modular design, where each educational feature is implemented as an independent Python module. This modular approach improves code readability, simplifies maintenance, and makes it easier to add new features in the future.

The backend includes modules such as:

- Question Answering Module
- Explanation Module
- Summary Module
- Quiz Generator Module
- Learning Path Module
- PDF Summarization Module

Each module generates a specific prompt based on user input and sends it to the Google Gemini API for processing.

TESTING:

The application was tested for:

- Question Answering
- Concept Explanation
- Text Summarization
- Quiz Generation
- Learning Path Recommendation
- PDF Upload
- Voice Input
- Text-to-Speech
- Download Functionality
- Responsive Design

All modules function correctly under normal operating conditions.

ADVANTAGES:

- Easy to use
- AI-powered learning
- Fast response generation
- Responsive interface
- Voice interaction
- PDF summarization
- Downloadable responses
- Personalized learning
- Interactive user experience

LIMITATIONS

- Requires an internet connection
- Depends on Google Gemini API availability
- Voice recognition depends on browser compatibility
- PDF summarization quality depends on document content

CONCLUSION:

EduGenie successfully demonstrates the practical application of Artificial Intelligence in education by integrating multiple learning services into a single platform. The system enables users to ask questions, understand concepts, summarize notes, generate quizzes, receive personalized learning recommendations, and summarize PDF documents efficiently. Features such as voice input, text-to-speech, downloadable responses, dark mode, and responsive design enhance usability and accessibility.

The project provides a scalable foundation for future AI-powered educational systems and highlights how modern AI technologies can improve the learning experience for students and educators alike.